

Smart AI-Based Multi-Payment Vending Machine using Verilog HDL

Project Report

1. Introduction

The Smart AI-Based Multi-Payment Vending Machine is a digital hardware system designed using Verilog HDL. The project simulates a modern vending machine capable of handling:

- Multiple product selection
- Multiple quantities
- Dynamic pricing
- Cash payment
- QR payment
- Fraud detection
- Timeout handling
- Recommendation engine
- Inventory management
- Transaction logging
- Admin control

The design follows a modular hardware architecture where each subsystem performs a dedicated task.

2. Objectives

The main objectives of the project are:

- To design a smart vending machine using Verilog HDL
- To support both cash and QR-based payments
- To implement inventory monitoring and stock verification
- To implement product recommendation
- To provide dynamic pricing based on demand
- To detect fraud and lock the machine
- To maintain transaction records

- To simulate real-world vending machine operations
-

3. Features of the System

Main Features

1. Multi Product Selection

Users can select multiple products simultaneously.

2. Multiple Quantity Handling

Each product can be purchased in different quantities.

3. Dynamic Pricing

Prices vary according to demand levels.

4. Recommendation Engine

The system recommends related products.

5. Cash Payment Support

Accepts coins and verifies payment.

6. QR Payment Support

Supports digital QR-based transactions.

7. Fraud Detection

Detects suspicious activities and locks the machine.

8. Timeout Handling

Cancels delayed or inactive transactions.

9. Inventory Management

Tracks available stock and updates inventory.

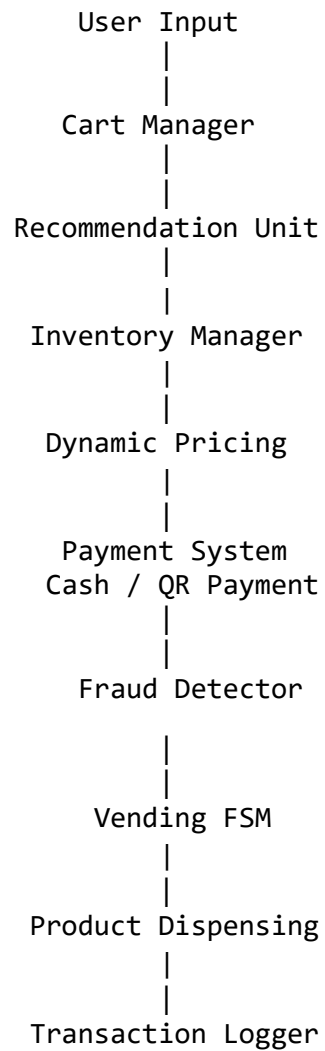
10. Transaction Logging

Stores transaction information for monitoring.

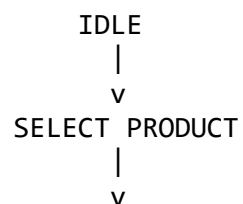
11. Admin Controller

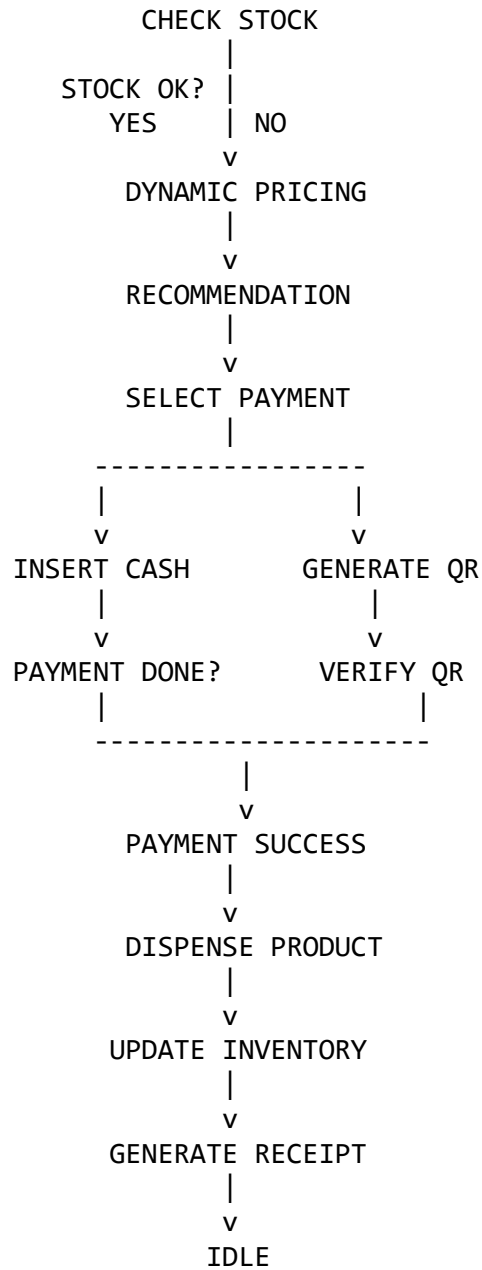
Allows admin unlock and maintenance.

4. Overall System Architecture



5. FSM Flowchart





6. Module Description

6.1 Vending FSM

Purpose

Controls the overall operation of the vending machine.

Functions

- Product selection
- Stock verification
- Payment handling
- Product dispensing
- Error handling
- Fraud locking

States Used

- IDLE
 - SELECT_PRODUCT
 - CHECK_STOCK
 - DYNAMIC_PRICING
 - RECOMMENDED_PRODUCT
 - SELECT_PAYMENT
 - INSERT_CASH
 - GENERATE_QR
 - WAIT_FOR_SCAN
 - VERIFY_PAYMENT
 - PAYMENT_SUCCESS
 - RETURN_CHANGE
 - DISPENSE_PRODUCT
 - UPDATE_INVENTORY
 - GENERATE_RECEIPT
 - TRANSACTION_LOG
 - UPDATE_HISTORY
 - CANCEL_TRANSACTION
 - TIMEOUT_STATE
 - LOCK_MACHINE
-

6.2 Cart Manager

Purpose

Stores and updates selected products.

Functions

- Adds user-selected products

- Updates quantities
 - Adds recommended products
-

6.3 Recommendation Engine

Purpose

Suggests products based on user cart.

Example

- If item0 is selected, item1 may be recommended.
- If item2 is selected, item3 may be recommended.

Benefits

- Increases sales
 - Simulates AI recommendation system
-

6.4 Inventory Manager

Purpose

Maintains stock levels.

Functions

- Checks stock availability
 - Updates inventory after dispensing
 - Prevents invalid purchases
-

6.5 Dynamic Pricing Module

Purpose

Adjusts prices according to demand.

Demand Levels

Demand Level	Pricing Action
Low	10% Discount
Normal	Original Price
High	10% Price Increase

6.6 Payment Controller

Purpose

Handles cash and QR payments.

Cash Payment Features

- Coin insertion
- Balance checking
- Change return

QR Payment Features

- QR verification
 - Digital payment support
-

6.7 QR Payment FSM

Purpose

Controls QR transaction sequence.

Operations

- QR activation
 - Payment validation
 - Timeout handling
 - Fraud handling
-

6.8 Fraud Detector

Purpose

Detects suspicious activities.

Example Conditions

- Invalid payment behavior
- Abnormal coin input
- QR timeout

Action

Locks the machine.

6.9 Timeout Controller

Purpose

Cancels inactive transactions.

Functions

- Prevents machine hanging
 - Improves reliability
-

6.10 Transaction Logger

Purpose

Maintains transaction history.

Stored Information

- Number of transactions
 - Last transaction amount
-

6.11 Admin Controller

Purpose

Provides maintenance control.

Functions

- Unlock machine after fraud
 - Enable maintenance/refill
-

7. Test Cases Implemented

Test Case	Description
TC1	Single product single quantity
TC2	Single product multiple quantity
TC3	Multiple products multiple quantities
TC4	Insufficient cash payment
TC5	QR payment success

Test Case	Description
TC6	QR timeout failure
TC7	Transaction cancellation
TC8	Fraud detection
TC9	Admin unlock
TC10	Out of stock condition
TC11	Recommendation acceptance
TC12	High demand dynamic pricing

8. Simulation Tools Used

Software

- Icarus Verilog (iverilog)
- GTKWave
- Visual Studio Code

Simulation Commands

Compile

```
iverilog -o sim tb/top_module_tb.v src/*.v
```

Run Simulation

```
vvp sim
```

Open Waveform

```
gtkwave wave.vcd
```

9. Waveform Analysis

The waveform verifies:

- FSM state transitions
- Payment success/failure
- Timeout conditions
- Fraud detection
- Inventory updates
- Product dispensing

- QR validation

Important signals observed:

- state
 - total_price
 - qr_done
 - cash_done
 - timeout
 - fraud_detected
 - stock_available
-

10. Advantages of the System

- Supports modern digital payments
 - Modular and scalable architecture
 - Realistic vending machine simulation
 - Improved customer interaction
 - Smart recommendation capability
 - Reliable fraud detection
 - Automated inventory tracking
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11. Applications

- Smart retail systems
 - Automated kiosks
 - Railway stations
 - Shopping malls
 - Airports
 - Educational FPGA projects
 - Embedded system research
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12. Future Enhancements

Future improvements may include:

- Real-time analytics

- Voice interaction
- Touchscreen interface
- OTP-based admin access

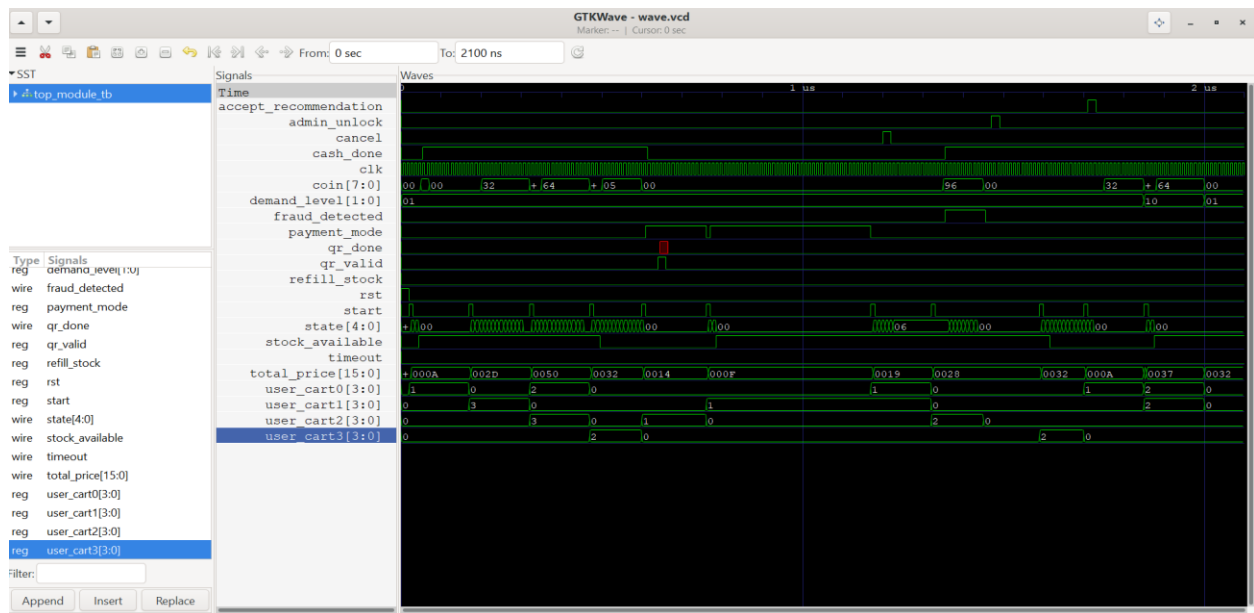
13. Conclusion

The Smart AI-Based Multi-Payment Vending Machine successfully demonstrates the implementation of a modern vending machine using Verilog HDL. The system integrates inventory management, digital payments, recommendation systems, fraud detection, and dynamic pricing into a single modular architecture.

The project provides strong understanding of:

- Finite State Machines
- Digital system design
- RTL modeling
- FPGA-oriented architecture
- Modular hardware development
- Simulation and verification

The project is scalable and can be further extended into a real FPGA-based smart vending solution.



```
PS C:\Users\bhavi\OneDrive\Desktop\iverilog\smart_vending_machine> iverilog -o sim tb/top_module_tb.v src/*.v
PS C:\Users\bhavi\OneDrive\Desktop\iverilog\smart_vending_machine> vvp sim
```

TESTCASE 1 : SINGLE PRODUCT SINGLE QUANTITY CASH SUCCESS

PAYMENT SUCCESSFUL
CHANGE RETURNED = 10

TESTCASE 2 : SINGLE PRODUCT MULTIPLE QUANTITY

DISPENSE SUCCESSFUL (ALL ITEMS)
=====

Thank you for purchasing!
Please visit again.
=====

TESTCASE 3 : MULTIPLE PRODUCTS MULTIPLE QUANTITIES

DISPENSE SUCCESSFUL (ALL ITEMS)
=====

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=====

TESTCASE 4 : INSUFFICIENT CASH PAYMENT

INSUFFICIENT STOCK
STOCK : 8 2 4 0
CART : 0 0 0 2
DISPENSE FAILED - INSUFFICIENT STOCK
STOCK: 8 2 4 0
CART : 0 0 0 2
=====

```
PS C:\Users\bhavi\OneDrive\Desktop\iverilog\smart_vending_machine> vvp sim
```

TESTCASE 5 : QR PAYMENT SUCCESS

QR PAYMENT SUCCESSFUL
TOTAL PRICE = 20
QR PAYMENT SUCCESSFUL
TOTAL PRICE = 20

TESTCASE 6 : QR TIMEOUT FAILURE

TESTCASE 7 : CANCEL TRANSACTION

TESTCASE 8 : FRAUD DETECTION

PAYMENT SUCCESSFUL
CHANGE RETURNED = 110
DISPENSE SUCCESSFUL (ALL ITEMS)
=====

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=====

TESTCASE 9 : ADMIN UNLOCK

TESTCASE 10 : OUT OF STOCK

INSUFFICIENT STOCK
STOCK : 8 2 2 0
CART : 0 0 0 2
DISPENSE FAILED - INSUFFICIENT STOCK

TEST CASES

```
TESTCASE 10 : OUT OF STOCK

INSUFFICIENT STOCK
STOCK : 8 2 2 0
CART : 0 0 0 2
DISPENSE FAILED - INSUFFICIENT STOCK
STOCK: 8 2 2 0
CART : 0 0 0 2
=====
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Please visit again.
=====

TESTCASE 11 : RECOMMENDATION ACCEPTED

=====
Thank you for purchasing!
Please visit again.
=====

TESTCASE 12 : HIGH DEMAND DYNAMIC PRICING

ALL TESTCASES COMPLETED

tb/top_module_tb.v:252: $finish called at 2100000 (1ps)
○ PS C:\Users\bhavi\OneDrive\Desktop\iverilog\smart_vending_machine> |
```

THANK YOU